

Afnan Alkhaibari

Academic & Research Portfolio

Master of Science (Bioinformatics) student
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Research direction

Interested in computational approaches to biological and clinical questions, particularly machine learning in health, genomics, and transcriptomic data analysis. Seeking a 2027 master's research project with opportunities to develop practical bioinformatics skills.

Education

Master of Science (Bioinformatics)

The University of Melbourne | 2025-present

Bachelor of Science in Biology

Prince Sattam bin Abdulaziz University, Saudi Arabia | 2024

Current collaborative research

Research team member | Ongoing

Participating formally in a multidisciplinary project involving disease-modifying treatments and immune-related disease topics. The team is currently collecting data from selected datasets and research publications.

- Contribute to the current data-collection phase within a 15-member team that includes doctors and pharmacists.
- Attend two to three research meetings each week to discuss progress and coordinate work.

Planned next phase: work with real patient data. The team also intends to publish its research process and subsequent findings; these activities are planned and are not completed outputs.

Selected Academic Work

01 / Machine learning in breast cancer

XGBoost for Breast Cancer Prediction

Coursework research plan

Developed a research plan exploring XGBoost classification across WDBC, METABRIC, and TCGA-BRCA datasets, with an emphasis on evaluation, interpretability, and reproducibility.

Work presented

- Motivated a research question through a review of breast-cancer prediction literature.
- Described preprocessing, handling of missing values and class imbalance, stratified train/test splits, and cross-validation.
- Outlined evaluation using ROC-AUC, accuracy, F1-score, and confusion matrices.
- Discussed SHAP-based model interpretation and a Galaxy Australia workflow for reproducibility.

What this demonstrates

Research planning, literature synthesis, familiarity with machine-learning evaluation concepts, and attention to model interpretation.

Scope and interpretation

This is a coursework research plan, not a published or clinically validated study. The datasets differ in features and prediction targets; separate within-dataset analyses do not establish transfer of a trained model to an external cohort. Reported pilot scores are therefore not presented here as verified evidence of cross-cohort generalisation.

Supporting work

Assignment 2 submission: "Use machine learning XGBoost to improve the prediction of breast cancer risk". Full coursework document available separately.

Genomics & Computing

02 / Critical appraisal

Chromosome-Level Genome Assembly Review

Coursework critical review | *Plagiognathops microlepis*

Critically reviewed a published genome-assembly study combining PacBio HiFi, Illumina, and Hi-C sequencing. Discussed assembly completeness, annotation, sampling limitations, and the interpretation of haplotype and mitochondrial information.

Skills demonstrated: scientific reading, methodological critique, interpretation of genome-assembly quality measures, and concise academic writing. Sequencing technologies and assembly tools were evaluated through the literature; this review does not establish hands-on use of those tools.

Source study: Wu et al. (2024), Scientific Data 11, 802.

DOI: 10.1038/s41597-024-03645-x

03 / Experimental computing

Array and Max-Heap Performance

Coursework experimental report | Python

Presented four experiments comparing array-based and max-heap behaviour: push-only workloads, changing getTop and pop frequencies, and bottom-up heap construction versus repeated insertion.

- Documented a Python environment using PyCharm, random, time, and matplotlib.
- Presented runtime plots and related observed differences to operation complexity and workload composition.
- Discussed how random inputs and implementation costs can affect observed performance relative to worst-case analysis.

Skills demonstrated: computational experimentation, performance visualisation, interpretation of algorithms, and technical reporting. The submitted report was reviewed for this portfolio; implementation code was not supplied.

Supporting work

Critical Review of the Chromosome-Level Genome Assembly of *Plagiognathops microlepis*; Report for assignment 2 (array and max-heap experiments). Full coursework documents available separately.